

34 Scientists compared the density of goblet cells in the lungs and the density of mucus in the lungs of three groups of people:

- people who do not smoke and do not have lung disease
- people who smoke tobacco but do not have lung disease
- people who smoke tobacco and have lung disease.

The results are shown in the table.

group	goblet cell density /cells per mm ²	mucus density /arbitrary units
non-smokers	19	6
smokers who do not have lung disease	54	26
smokers with lung disease	37	15

What is indicated by these data?

- 1 There is a positive correlation (relationship) between density of goblet cells and density of mucus.
- 2 Lung disease results in an increase in goblet cell density.
- 3 There is an association between tobacco smoking and an increase in mucus density.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only